$$= \cos u \times (2x)$$
$$= \cos(x^2).2x$$
$$= 2x\cos(x^2).$$

(ii) $u = \sin x$

Then, $y = u^2$

and $\dfrac{dy}{dx} = \dfrac{dy}{du} \times \dfrac{du}{dx}$

$$= 2u \times \cos x$$
$$= 2 \sin x \,.\, \cos x$$
$$= \sin 2x.$$

Example 10.10

Differentiate : $y = (x^3 - 1)^{100}$.

Solution

Take $u = x^3 - 1$ so that

$y = u^{100}$

and $\dfrac{dy}{dx} = \dfrac{dy}{du} \times \dfrac{du}{dx}$

$$= 100u^{100-1} \times (3x^2 - 0)$$
$$= 100(x^3 - 1)^{99} \times 3x^2$$
$$= 300x^2(x^3 - 1)^{99}.$$

Example 10.11

Find $f'(x)$ if $f(x) = \dfrac{1}{\sqrt[3]{x^2 + x + 1}}$.

Solution

First we write : $f(x) = (x^2 + x + 1)^{\frac{-1}{3}}$

Then, $f'(x) = -\dfrac{1}{3}(x^2 + x + 1)^{\frac{-1}{3}-1} \dfrac{d}{dx}(x^2 + x + 1)$

$$= -\frac{1}{3}(x^2 + x + 1)^{\frac{-4}{3}} \times (2x + 1)$$
$$= -\frac{1}{3}(2x + 1)(x^2 + x + 1)^{\frac{-4}{3}}.$$

Example 10.12

Find the derivative of the function $g(t) = \left(\dfrac{t-2}{2t+1}\right)^9$.

Solution

$$g'(t) = 9\left(\frac{t-2}{2t+1}\right)^{9-1} \frac{d}{dt}\left(\frac{t-2}{2t+1}\right)$$

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